

CHE 363 Separation Processes and Mass Transfer (Unique No 16585) Spring 2026 Syllabus

Lectures: T Th 11:00 AM – 12:30 PM, CPE 2.218

Recitation: F 1 PM – 2 PM, UTC 4.132

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1. Course content (from course catalog)

Design and analysis of equilibrium and mass transfer-based separations such as absorption, chromatography, crystallization, distillation, extraction, and membrane-based processes. Three lecture hours and one recitation hour a week for one semester. Chemical engineering majors must make a grade of at least C in this course. Prerequisite: Chemical Engineering 322, 348, and 353 with a grade of at least C in each.

Chapters (McCabe et al. (6th or 7th edition required- very similar)

UNIT 1:

Chapter 17 Diffusion and Mass Transfer between Phases (steady and unsteady state) ~3 ½ weeks

First Exam

UNIT 2:

Chapter 18 Gas Absorption (excluding rich gases and chemical reaction) 1.5

Chapter 20 Equilibrium stage operations 1/2

Chapter 21 Distillation 2

Second Exam

UNIT 3:

Chapter 25 Adsorption 2

Chapter 26 Membrane separations; gases, liquids- osmosis and reverse osmosis 2

Third Exam

Final Exam (Cumulative)

2. Course objectives and outcomes

After taking this course, you should be able to

- Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- Engineering design for open ended problems: Synthesize material balances, thermodynamics and mass transfer to do stagewise calculations
- Define characteristics of the major separations processes used by chemical engineers
- Identify an appropriate separation to use given a situation
- Perform Thermodynamic calculations in distillation and stage efficiencies
- Analyze mass transfer in equilibrium and rate-based as well as differential and staged separations
- Design separation columns using both graphical techniques, analytical models and process simulation models (i.e., Aspen).
- Calculate overall mass transfer coefficients across multiple layers of resistance, based on gas or liquid layer
- Predict mass transfer coefficients for particular geometries based on correlations
- Identify dominant resistance layer limiting overall mass transfer and thus quality of the separation
- Design tower height and diameters for both equilibrium and rate-based separations.
- Given a tower height, calculate component concentrations in exit streams.
- Given separation specifications, calculate liquid/ gas flow-rates.

3. Prerequisites

This course *will use and build* upon concepts learned in previous courses

- Diagram, set up and solve mass and energy balances (Che 317)
- Calculate vapor-liquid and liquid-liquid equilibrium (Che 322)
- Apply Raoult's Law to calculate partial pressures (Che 322)

4. Course materials

We will be using the following book: Unit Operations of Chemical Engineering by Julian Cleveland Smith, Peter Harriott, and Warren L. McCabe

Some other related references in the library are:

Cussler "Diffusion: Mass Transfer in Fluid Systems (non-required suppl. text – key info in sup. Notes in Canvas)

Seider, Henley, Roper, Separation Process Principles: With Applications Using Process Simulators (non-required suppl. text – key info in sup. Notes in Canvas)

Longhorn Textbook Access

The materials for this class are available through the Longhorn Textbook Access (LTA) program, a collaboration between UT Austin, The University Co-op and textbook publishers to significantly reduce the cost of digital course materials for students. You can access your required materials through the “My Textbooks” tab in Canvas. You are automatically opted into the program but can easily opt-out (and back in) via Canvas through the 12th class day. If you remain opted-in at the end of the add/drop period (12th class day fall/spring, 4th class day summer sessions), you will receive a bill through your “What I Owe” page. If you do not pay your bill by the specified deadline, you will lose access to the course materials and your charge will be removed. More information about the LTA program is available at the Longhorn Textbook Access website.

5. Structure of the course

5.1 Lectures and recitations

The lectures will introduce key theoretical concepts, which will be supported with some examples. Further examples and more complex problems will be discussed during the recitation period. Reading assignments will be given (typically together with the homework) and will address the topics to be covered in lectures in the near future. **Attendance to both lectures and recitations is highly encouraged.** There is a strong correlation between class attendance and participation in course activities and understanding/performance in the course, therefore, I strongly advise you to attend lectures. Please be courteous to others during class: I typically don’t want you speaking with each other during class. Thank you in advance!

Aspen software sessions will be in recitation only.

I can assign $\pm$ points to your final average based on your professionalism and participation. For example, if you are consistently wide-awake in class asking good questions during every class period, answering the questions I pose to the class, and arriving in class ready to go before the bell rings, I’ll add points to your final average at the end of the semester.

5.2 Cell phones/laptops

Please don’t use cell phones or laptops (or any electronic device) during class unless it is for taking notes. Please turn them off (or put them in silent mode) and stow them away before class begins. Although these electronic devices can be quite useful, they can also be very distracting to you, other students, and me. Your cooperation is greatly appreciated!

6. Homework

Homework problems will be assigned once a week. Unless otherwise stated, the assignment will be due in exactly one week by 11:59PM. **Homeworks submitted up to 24 hours late will get 60% credit, then no credit after that.** The name of the student and EID number should be clearly written on the first page of the homework. The homework will be submitted online via Canvas.

You are encouraged to discuss the material of the course, but when it comes to assigned work, **you are expected to do it independently**. Copying or paraphrasing from the web, another source, or another student's solution or permitting your own solution to be copied or paraphrased is considered cheating. In any case of plagiarism or cheating, zero credit will be given for the work in question, and in some circumstances, an F may be granted for the entire course.

7. Examinations

There will be three midterm exams and a final exam. **All exams will be closed book and closed notes. A sheet with Mass transfer summary and equations needed will be provided. The exams will be held during the class (~1.5 hrs).** The midterm exams will test everything covered in lectures, recitations and reading assignments, from the beginning of the course until the exam date. Questions for each unit exam (~20) will be directly from a question bank and could include questions from HW or in class problems. 85% of the grade (total points in each exam) will be from these questions. Some additional sections will be added to these questions (for example an additional sub question for a main problem). Final exams will include all question bank problems, HWs, and in class problems

Communication devices (cell phones, etc.) are not permitted during the exams. No makeup exams will be given. Students who miss an exam will be given a zero grade for the exam. The exam can be taken at a later time only under the circumstances presented in the Accommodations, Absences, and Religious Holidays sections.

The exam dates are

- Exam 1: February 12
- Exam 2: March 12
- Exam 3: April 16
- Final: May 2, 8:00 AM - 10:00 AM (location TBD)

8. Web site information and communication

For course information, announcements, homework assignments, and solutions, you are strongly encouraged to visit the course's website on Canvas regularly. Homework solutions will be posted shortly after the homework is due. Please ensure that your notification settings alert you about new announcements. In addition, students are encouraged to post both conceptual and problem-set questions to the canvas discussions tab. Both asking and answering good questions in the discussions tab and during lectures are highly encouraged. If you email the instructor or teaching assistant, please include [CHE363S26] (no spaces, with square brackets) in the subject line so that the message is sorted appropriately.

9. Grading

The weighted average grade will be computed by considering the exams, homework assignments, and an aspen project as follows:

- Exam 1: 20 %
- Exam 2: 20 %
- Exam 3: 20 %
- Homework: 10 %
- Final exam: 25 %
- Aspen Project: 5%

Requests for regrades on tests or assignments will be accepted if submitted to Prof. Mitrai **within three days** after the test or assignment is returned. In such a case you should submit the original and a separate signed page describing exactly what is disputed.

Letter grades will be allocated based on the scaled scores presented in table below

From	To	Letter grade
0	59.99	F
60	62.99	D-
63	65.99	D
66	69.99	D+
70	72.99	C-
73	75.99	C
76	79.99	C+
80	82.99	B-
83	85.99	B
86	89.99	B+
90	92.99	A-
93	100	A

10. Accommodations

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <http://disability.utexas.edu/>. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

11. Absences

Students must provide a certified medical excuse letter or a letter on University letterhead (for school-related activities) in order to obtain reschedules of missed exams.

12. Religious holiday

By UT Austin policy, you must notify the faculty (Prof. Mitrai) of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

13. Academic integrity

Academic integrity violations will result in a 0 (zero) grade for the respective exam or assignment, and an “F” grade for the entire course. Forms of scholastic dishonesty include (but are not limited to) plagiarism, use of unauthorized materials for exams, and changing solutions to returned assignments, and exams. Further information is available at (link last accessed 8/23/2024):\\ <https://deanofstudents.utexas.edu/conduct/academicintegrity.php>

14. Sharing course materials is prohibited

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University’s Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

15. How to succeed in this class

- **Diagram the problem** - The math is not extremely hard (mostly algebra and separable ODE’s), but what can be hard is envisioning the physical situation – I always draw pictures to identify the direction of mass transfer, the layers of resistance, etc.
- **Do the homework** – try to understand the steps.
- **Do more than the HW** – look over and try to understand your notes, read the text, explain things to your roommate – whatever works for you!
- Review info where you first learned it. Reread key sections in calculus book, transport book, thermo book etc. and your old notes. The more you review these topics the better you will retain it.

- The textbook is a key element for this class. As you do lifelong learning, reading books and articles play an important role. **The notes are not meant to be self-sufficient, but to complement the text. Read them together!! The notes refer to text locations.**
- See us in office hours. Participate in lecture and recitation.
- **It usually takes ~ 8 hours/week outside of class to read the book including examples in chapters (in conjunction with notes), learn the theory and concepts, and to do the homework problems.**
- Strive to understand the theory and concepts in addition to doing problems.
- Learn to identify classes of problems based on theoretical concepts. The summary sheets called exam study guide in the chapters will help here.
- Maximize what you get out of the lectures by keeping up and understanding all concepts after each lecture. If you get behind, try to immediately catch up to maximize efficiency.
- Look ahead in the book and notes for about 5-10 min before each lecture. The brain will then be activated and eager to learn very efficiently and deeply.
- Avoid the short cut to use old homework solutions. It is far easier to follow somebody else's solution than to construct your own.
- Always check units in equations. They are important.
- Do all homework problems and study solutions after turning in homework as needed on Canvas. Save hws when you get them back so you can study your approach vs. posted solutions.

16. LearnChemE screencasts <http://www.learncheme.com>.

All of you learn differently. We wish to reach everyone. These highly acclaimed optional screencasts for PCs to tablets to phones may be very helpful to many of you.

17. TITLE IX DISCLOSURE

Texas Education Code, Section 51.252 requires all employees of Texas universities, including faculty, to report to the Title IX Office any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this nature (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with any faculty or staff member about an incident of this nature, please be aware that they are required to report that information.

Graduate teaching and research assistants are also designated as mandatory reporters and are required to report behaviors that are considered sexual misconduct, including the types of misconduct described above.

Pregnant and parenting students are also eligible to receive support and resources from the Student Parent Liaison in the Title IX Office . Support and resources may include supportive measures, medically necessary periods of leave, and leaves of absence. Case managers from the Title IX Office connect with students to provide one-on-one, individualized support. If you would like to speak with a case manager for support, please email: supportandresources@austin.utexas.edu. For more information about reporting options and resources, please visit the Title IX website, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.